

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination

University Theory Examination November 2017

PH205 Pharmaceutical Engineering - III

Date: 29.11.2017, Wednesday

Time: 10:00a.m. to 1:00p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION – I

- Q 1 Attempt all questions. Each question is of one mark. 20
1. _____ is designed to streamline and improve the accuracy and efficiency of the medication use process.
[A] Machinery
[B] Automation
[C] CPU
[D] Controller
 2. Computer process control does not measure _____.
[A] Temperature
[B] Flow rate
[C] Pressure
[D] Personal error
 3. In more volatile concentrated solutions, as distance increases the concentration _____.
[A] increases.
[B] decreases.
[C] remain constant.
[D] None of the above.
 4. Piezo meter is used for the measurement of _____.
[A] Pressure
[B] Voltage
[C] Temperature
[D] None of the above
 5. As per BP specification, moderately fine powder pass through ____ no. of sieve.
[A] 44
[B] 11
[C] 22
[D] 88
 6. Fluid energy mill is used for _____.
[A] Suspension
[B] Emulsion
[C] Both

- [D] None of the above
7. By sedimentation and elutriation, _____ micrometer particle size can be determined.
 [A] 2 – 200 micrometer
 [B] 200 – 400 micrometer
 [C] 2 – 50 micrometer
 [D] 1 – 10 micrometer
 8. Mechanism of fluid energy mill is _____
 [A] Attrition
 [B] Impact
 [C] Both
 [D] None of the above
 9. 5 tones = _____ lb (pound)
 [A] 11200
 [B] 1120
 [C] 11.20
 [D] 112
 10. Approximately screening area of the sieve is _____ % of the total area.
 [A] 35-40
 [B] 25-30
 [C] 10-15
 [D] None of the above
 11. One of the following is not used for size reduction.
 [A] Comminution
 [B] Grinding
 [C] Shaking
 [D] Cutting
 12. Quasi emulsion solvent diffusion method is used for preparation of _____
 [A] Spherical crystal
 [B] Needle crystal
 [C] Plate crystal
 [D] Hexagonal crystal
 13. Squirrel cage disintegrator is also known as _____
 [A] Cage mill
 [B] Bucket mill
 [C] Bull mill
 [D] Hanging mill
 14. A process of crystallization taking place from vapor of a substance is known as
 [A] Sublimation
 [B] Precipitation
 [C] Filtration
 [D] Distillation
 15. 1520 mm Hg = _____ bar
 [A] 4.052
 [B] 2026
 [C] 2.026
 [D] 8.932
 16. Which of the following is the example of crystal habit
 [A] Plate
 [B] Blade
 [C] Both
 [D] Hexagonal
 17. Knudsen is one type of vacuum _____ gauge.

- [A] Diaphragm
 [B] Ionization
 [C] Radiometer
 [D] Compression
18. In End runner mill, crushing is done due to heavy weight of _____
 [A] Steel pestle
 [B] Steel mortar
 [C] Steel bowl
 [D] None of the above
19. 5 kg = _____ lb (pound)
 [A] 12.32
 [B] 11.025
 [C] 500
 [D] 50
20. DDC (used in computer) means
 [A] Direct digital controller
 [B] Direct dust controller
 [C] Dust digital controller
 [D] None of the above

SECTION – II

- Q 2 Attempt any **TWO** of the following
- A Write a shortnote on Swenson walker. 05
- B Explain in brief about the factors affecting crystallization. 05
- C Write construction and working of vacuum crystallizer with neat and clean diagram. 05
- Q 3 Attempt any **TWO** of the following
- A Write a shortnote on solid solid mixing. 05
- B How the content uniformity can be maintained through mixing? 05
- C Write the importance of automatic process control in pharmaceutical industry. 05

SECTION – III

- Q 4 Attempt any **FOUR** of the following
- A What is comminution? Write a brief note on Ball mill. 05
- B What are the objectives of size reduction? Explain with proper examples. 05
- C Discuss factors affecting milling of pharmaceutical substances. 05
- D Explain elutriation. Differentiate dry and wet milling with proper examples. 05
- E Write a detail note on precautions to be taken for preventing hazards in pharmaceutical industries. 05
- F What is cyclone separator? Explain principle, working and construction of the same. 05
- Q 5 Attempt any **FOUR** of the following
- A Discuss the theory of milling. 05
- B Write a note on chemical hazards citing ralted incidences. 05
- C Discuss any one mill used for preparation of emulsion in detail. 05
- D Write a short note on sieve shaker. 05
- E How abrasive materials are reduced in size? Explain suitable mill for the same. 05
- F Write a note on particle engineering. How this concept can be helpful in pharmaceutical field? 05